



SK Saif Ibna Ezhar Arko

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

Research Interests

Quantum Information Theory • Quantum Computing and Algorithms • Quantum Cryptography • Quantum Machine Learning • Adversarial Robustness of Quantum Models • Post-Quantum Cryptography.

Education

BRAC University

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

2026

CGPA: 3.20 / 4.00

- **Undergraduate Thesis:** *Quantum-Aware Image Encoding and Adversarial Perturbation*
- **Relevant Coursework:** Quantum Computing, Quantum Error Correction, Quantum Algorithms, Machine Learning, Pattern Recognition, Data Structures & Algorithms, Operating Systems, Computer Graphics, Linear Algebra, Probability & Statistics, Discrete Mathematics.

Notre Dame College, Dhaka

Dhaka, Bangladesh

Higher Secondary Certificate (Science)

2020

GPA: 5.00 / 5.00

St. Gregory's High School and College

Dhaka, Bangladesh

Secondary School Certificate (Science)

2018

GPA: 5.00 / 5.00

Research Experience

Undergraduate Thesis Researcher

2024 – Present

BRAC University, Department of Computer Science and Engineering

Supervisor: Ipshita Bonhi Upoma, Lecturer, Department of CSE, BRAC University.

- Developing theoretical frameworks for encoding classical image data into quantum states using amplitude and basis encoding schemes.
- Investigating the mathematical robustness of quantum image representations against adversarial noise and perturbations.
- Analyzing the trade-offs between quantum encoding fidelity, circuit depth, and resilience to bit-flip and phase-flip channels.

Student Researcher

September 2026 – January 2027

ELITE Research Lab LLC — Research Engagement

Supervisor: Md Kishor Morol

- Conducting research at the intersection of quantum computing and machine learning in collaboration with industry experts.
- Contributing to ongoing research projects and exploring innovative developments across quantum and AI/ML domains.

Research Intern, QIntern 2026 Global Internship

Summer 2026

QWorld — International Quantum Computing Organization

Supervisor: Dr. Maha A. Metawei

- Project: *Accelerating hybrid quantum classical machine learning tasks on HPC platforms (QNLP).*

- Implemented high-performance quantum circuit simulations and tensor network optimizations to scale hybrid algorithms on distributed clusters.
- Extending core framework classes like *Ansatz* and *Reader* to customize the QNLP pipeline, while developing unified data format conversion scripts.
- Collaborated with an international team of researchers across multiple countries.

Research Intern, QIntern 2025 Global Internship

Summer 2025

QWorld — International Quantum Computing Organization

Supervisor: Dr. Kumar Prateek, Dr. Soumyadev Maity

- Project: *Design of Quantum-Anamorphic Encryption Against Coercive Surveillance.*
- Researched and prototyped quantum-anamorphic encryption schemes to provide deniability and resistance against coercive adversaries.
- Collaborated with an international team of researchers across multiple countries.
- **Achievement:** Project awarded **3rd Best Project globally** among submissions worldwide.

Selected Academic Projects

Monkeypox Classification Using a Hybrid Quantum CNN (HQCNN)

[\[GitHub\]](#)

- Designed a Hybrid Quantum Convolutional Neural Network combining classical feature extraction with parameterized quantum circuits.
- Benchmarked the hybrid model against purely classical CNN architectures for medical image diagnosis.

QVTM — Quantum Variable Topic Model (Global Quantum Hackathon 2024)

[\[GitHub\]](#)

- Implemented a Quantum Variable Topic Model leveraging variational quantum circuits for unsupervised topic discovery, developed for the **Global Quantum Hackathon 2024**.
- Designed an encoder-decoder architecture employing **amplitude embedding** to represent classical text data within the quantum Hilbert space.
- Optimized the model using a composite objective combining **KL-divergence**, **Cross-Entropy**, and **Maximum Mean Discrepancy (MMD)** loss functions.
- Achieved superior performance with a **Topic Diversity score of 1.0** and improved coherence over traditional Latent Dirichlet Allocation (LDA) baselines.
- Developed entirely in Python (Jupyter Notebook) using a quantum machine learning stack.

Monkeypox Detection Using EfficientNetB3

[\[GitHub\]](#)

- Built a deep transfer-learning pipeline with EfficientNetB3 to classify Monkeypox from skin lesion images.
- Applied data augmentation and regularization to mitigate overfitting on a limited medical dataset.

NLP Similarity Classifier — Bidirectional LSTM Siamese Network with Multiple Merge Strategies

[\[GitHub\]](#)

- Designed a Bidirectional LSTM-based Siamese Network in TensorFlow/Keras for semantic textual similarity and paraphrase identification.
- Engineered a hybrid feature pipeline that fuses deep learning sentence embeddings with traditional NLP metrics such as Jaccard similarity and fuzzy string matching.
- Implemented and compared four distinct merge strategies (concatenation, absolute difference, element-wise product, and combined fusion) for effectively comparing question representations.
- Built an automated data augmentation pipeline generating over **7,400 synthetic samples** and applied class weighting to address dataset imbalance.
- Incorporated Early Stopping and model checkpointing to prevent overfitting and persist the best-performing model version.
- Developed a comprehensive evaluation suite featuring ROC curves, confusion matrices, and feature importance analysis, demonstrating high confidence in detecting paraphrased content and accurately filtering unrelated queries.

Time-Series Forecasting of Stock Prices — ARIMA vs. LSTM

[\[GitHub\]](#)

- Conducted a comparative analysis between classical statistical (ARIMA) and deep learning (LSTM) models for financial time-series forecasting.

SmartCare: Healthcare Management Web Application

[GitHub]

- Developed a full-stack healthcare management platform using Django with patient interaction, service management, and dynamic interfaces.

3D Ball Roller Game (Python + OpenGL)

[GitHub]

- Built a physics-based 3D game from scratch using legacy OpenGL with collision detection and camera controls.

Honors and Awards

- **3rd Best Project Globally** — QIntern 2025, QWorld International Internship Program (2025).
- **Diploma in Quantum Anamorphic Encryption** — QWorld QIntern 2025.

Professional Experience

Datasoft Systems Bangladesh Limited

Dhaka, Bangladesh

Software Developer(AI and Quantum Department)

December 2025 – Present

- Engineer production-ready, Python-based AI solutions targeting high-efficiency data analysis, document processing, and workflow automation for enterprise and financial application workloads.
- Contributed to the terminal operating system of Chittagong Sea Port, Bangladesh.
- Deploy and operate self-hosted LLM inference infrastructure with `llama.cpp/llama-server`, serving a 228B-parameter (10B active) Mixture-of-Experts model in quantized GGUF format across heterogeneous single-GPU nodes, sized for 10–20 bursty concurrent users.
- Designed the *Hermes Agent* framework: model-switching, multi-provider LLM routing (NVIDIA NIM, DeepSeek, OpenRouter, local Ollama) exposed through an authenticated API gateway running as a supervised `systemd` service on a shared multi-user server.
- Built LangGraph-based agentic SQA workflows that plan and execute multi-step verification tasks across Django/React/PostgreSQL projects, translating engineering domain knowledge into reusable AI automation.
- Delivered *iSAGE*, an internal assistant UI over the local LLM backend, integrating a bearer-authenticated inference API and a modular skills layer for task-specific agent capabilities.
- Design and maintain distributed web scraping pipelines and reliable ETL workflows supporting large-scale data collection, plus reproducible ingestion and preprocessing frameworks for downstream ML/DL models.
- Collaborate in a fast-paced environment applying Agile/Scrum methodologies, Git version control, and rigorous code reviews.

Maxcode Lab

Dhaka, Bangladesh

Python Developer Intern

October 2025 – December 2025

- Engineered Python tools for web scraping, automation, database management, and REST API integration.
- Developed scripts to extract, clean, and store structured data from heterogeneous online sources.

Teaching and Leadership Experience

Caretutors Technology Limited

Dhaka, Bangladesh

Verified Tutor

December 2024 – Present

- Deliver online and in-person Python instruction covering programming logic, data structures, and core language concepts.
- Design personalized, project-based learning plans aligned with each student's background and goals.

Community Contributions and Professional Service

IBM Qiskit Advocate

IBM Quantum

Advocate #129943

2025 – Present

- Recognized by IBM Quantum as a Qiskit Advocate for sustained contributions to the global quantum computing community through education, mentoring, and content creation.
- Lead quantum computing education initiatives and workshops for students and early-career researchers.
- Mentor aspiring quantum developers and researchers on Qiskit-based projects and quantum algorithms.
- Create and share quantum computing tutorials, articles, and learning resources for the wider community.
- Support the Qiskit community on official Discord and GitHub channels by answering questions and reviewing contributions.
- Organize and coordinate quantum computing events, hackathons, and outreach activities.
- Contribute to ongoing quantum computing research and innovation efforts within the Qiskit ecosystem.

QBangladesh — National Hub of QWorld

Founder, Coordinator, and Mentor

Bangladesh

April 2025 – Present

- Established and lead the first national quantum computing community in Bangladesh under the QWorld umbrella, building a sustainable ecosystem for quantum education and research.
- Organize and deliver hands-on workshops on quantum computing using Python, Qiskit, and Cirq for undergraduate and graduate students nationwide.
- Mentor early-career researchers on quantum algorithms, simulation, and applied quantum research.

Technical Skills

Programming Languages:	Python, C, C++, SQL
Quantum Frameworks and Tools:	Qiskit, Cirq, PennyLane (basic), Stim
ML / DL Frameworks:	TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib
Web & Graphics:	Django, HTML/CSS, OpenGL
Tools & Platforms:	Git, GitHub, Linux, Jupyter, LaTeX
Research Areas:	Quantum Algorithms, Quantum Cryptography, Quantum ML, Adversarial ML, Deep Learning

Certifications and Workshops

- **Quantum AI Lab – Quantum Cryptography** — Bibliotheca Alexandrina, 2026.
- **Quantum Business Foundations** — IBM, 2026.
- **Basics of Quantum Information** — IBM, 2026.
- **Pan-African Quantum Challenge 2025** — Africa Quantum Consortium, 2026.
- **Quantum Random Number Generation Certificate** — Quantum Security Defence, 2026.
- **Diploma: Design of Quantum-Anamorphic Encryption Against Coercive Surveillance** — QWorld, 2025.
- **Third Team Award: Design of Quantum-Anamorphic Encryption Against Coercive Surveillance** — QWorld, 2025.
- **Introduction to Quantum Computing** — Qubit by Qubit, Frontier Technology Institute, 2025.
- **QCOURSE 501-2: Elements of Quantum Computing and Programming** — QWorld, 2025.
- **QWorld’s QSilver26 Workshop** — QWorld, 2024.
- **QWorld’s Qbronze144 Workshop: Quantum Computing & Programming** — QWorld, 2024.
- **Quantum Machine Learning Fundamentals Course** — Ingenii, 2024.

Languages

- **Bengali** (Native)
- **English** (Full Professional Proficiency)

References

Ipshita Bonhi Upoma

Lecturer

Department of CSE

School of Data and Sciences

BRAC University, Bangladesh

(Undergraduate Thesis Supervisor)

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Dr. Maha Metawei

Researcher

HPC and Big Data Lab

Computer & Systems Department

Electronics Research Institute

New Nozha, Egypt

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Dr. Kumar Prateek

Assistant Professor (Grade-II)

Dept. of Computer Sc. & Engg.

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